

Café Sales Analysis

How the data was taken, cleaned and presented, step by step

1. Overview

This project looks at 160 café orders placed in March 2026 across five cities: Bengaluru, Chennai, Delhi, Mumbai and Pune. The aim was to answer three questions: which city earns the most, how much money is lost to cancelled and refunded orders, and which payment method customers prefer. Everything was done in Excel with formulas, so every number in the report can be traced back to the order data.

2. The data

The starting point was a list of café order records with 170 rows. Each row is one order. The raw columns were:

Column	What it holds
Order ID	Order number, such as ORD2001
Date	Order date (1 to 30 March 2026)
Customer	Customer name, typed in inconsistently
Product / Category	Item sold; category is Beverage, Bakery or Food
Quantity, Price (₹)	Units ordered and price per unit
Payment Method	Card, Cash or UPI
City	City of the order, with spacing and capital-letter mistakes
Order Status	Completed, Cancelled or Refunded, also with typing mistakes

Before changing anything I scrolled through the data to see what one row means and which column tells me whether money was actually kept. That column is Order Status: only Completed orders are money the café keeps.

3. Cleaning the data

Step 1: Remove duplicates

The same order appeared more than once. I used Data > Remove Duplicates on Order ID. This removed 10 rows and left 160 unique orders.

Step 2: Keep the original messy entries

The typed-in customer, city and status values were moved to grey columns P, Q and R on the Orders Data sheet. They are kept for reference and the cleaned versions are calculated from them.

Step 3: Fix spelling, spacing and capital letters

Entries like " mumbai", "DELHI " and "refunded " would be counted as different cities and statuses. Three formulas fix this (shown for row 2):

Cleaned column	Formula	Why
Customer (C)	=TRIM(PROPER(P2))	Removes extra spaces and fixes capital letters

Cleaned column	Formula	Why
City (I)	=TRIM(PROPER(Q2))	Makes " mumbai" and "DELHI " match "Mumbai" and "Delhi"
Order Status (J)	=TRIM(PROPER(R2))	Makes "refunded " match "Refunded"

Step 4: Flag orders with missing or invalid values

21 orders had a missing quantity, a zero or negative quantity, or a missing price. They cannot produce a sensible sales figure, so I flagged them in a Data Issue column (N):

Formula (row 2)
=IF(F2="", "Quantity missing", IF(F2<=0, "Quantity invalid", IF(G2="", "Price missing", "")))

These orders stay in the data but count as ₹0 sales, and they are highlighted red with conditional formatting.

4. Adding calculated columns

Three more columns were added on Orders Data so the summary tables could be built with simple formulas:

Column	Formula (row 2)	Meaning
Sales Value (K)	=IF(N2="", F2*G2, 0)	Quantity × Price, or 0 when there is a data issue
Order Size (L)	=IF(N2<> "", "", IF(F2>=3, "Bulk (3+ items)", "Regular"))	Bulk for 3 or more items, otherwise Regular
Counted as a Sale? (M)	=IF(J2="Completed", "Yes", "No")	Yes only for Completed orders

5. Building the summary

Every table on the Summary sheet is made with SUMIFS and COUNTIFS pointing at Orders Data (rows 2 to 161). Nothing is typed in by hand.

Step 1: Key numbers

Measure	Formula	Result
Total orders	=COUNTA('Orders Data'!\$A\$2:\$A\$161)	160
Total sales	=SUM('Orders Data'!\$K\$2:\$K\$161)	₹46,500
Sales kept	=SUMIFS('Orders Data'!\$K\$2:\$K\$161, 'Orders Data'!\$J\$2:\$J\$161, "Completed")	₹32,990
Completion rate	=COUNTIFS('Orders Data'!\$J\$2:\$J\$161, "Completed")/A6	70.6%
Sales lost	=C6-E6	₹13,510
Avg. completed order	=E6/COUNTIFS(Status, "Completed", Data Issue, "")	₹329.90

Step 2: Sales by city

For each city, total sales, sales kept, sales lost, the share lost and a rank are calculated (formulas for Bengaluru, row 64):

Column	Formula
Total sales	=SUMIFS('Orders Data'!\$K\$2:\$K\$161,'Orders Data'!\$I\$2:\$I\$161,A64)
Sales kept	=SUMIFS('Orders Data'!\$K\$2:\$K\$161,'Orders Data'!\$I\$2:\$I\$161,A64,'Orders Data'!\$J\$2:\$J\$161,"Completed")
Sales lost	=B64-C64
% of sales lost	=IF(B64=0,0,D64/B64)
Rank	=RANK(C64,\$C\$64:\$C\$68)

City	Total sales	Kept	Lost	% lost	Rank
Bengaluru	₹10,100	₹7,280	₹2,820	27.9%	3
Chennai	₹6,220	₹2,600	₹3,620	58.2%	5
Delhi	₹9,290	₹6,830	₹2,460	26.5%	4
Mumbai	₹10,980	₹7,830	₹3,150	28.7%	2
Pune	₹9,910	₹8,450	₹1,460	14.7%	1
Total	₹46,500	₹32,990	₹13,510	29.1%	

Step 3: Orders by status, category and payment method

Table	Formula pattern
Orders by status	=COUNTIFS('Orders Data'!\$J\$2:\$J\$161,H64) and =I64/\$A\$6 for the percentage
Sales by category	=SUMIFS(Sales Value, Category, A73) and with Status = "Completed" for kept sales
Payment methods	=COUNTIFS(Payment, H73, Data Issue, "") for valid orders, =SUMIFS(Sales Value, Payment, H73) for sales

Step 4: City by status table

The City x Status sheet works like a pivot table, but with formulas so it updates automatically. Each cell uses one SUMIFS with two conditions, city from the row and status from the column header:

Formula (Bengaluru, Cancelled)
=SUMIFS('Orders Data'!\$K\$2:\$K\$161,'Orders Data'!\$I\$2:\$I\$161,\$A5,'Orders Data'!\$J\$2:\$J\$161,B\$4)

Step 5: Check cells

To make sure nothing was missed, the totals are compared with the key numbers. For example, the city table shows a tick when its total equals total sales:

Formula
=IF(AND(B69=\$C\$6,C69=\$E\$6),"✓ Totals match the Key Numbers above","✗ Totals do not match - check the data")

Step 6: Findings text

The findings on the Summary sheet are written by formulas too, using INDEX and MATCH to pick the top city, so the wording follows the data. For example, the best city is found with:

Formula

=INDEX(\$A\$64:\$A\$68,MATCH(MAX(\$C\$64:\$C\$68),\$C\$64:\$C\$68,0))

6. Showing the data

The Summary sheet has four charts built from the tables above: sales by city, total vs. sales kept (bar); share of orders by payment method (pie); orders by status (bar); and sales by product category, total vs. sales kept (bar). Charts read straight from the summary tables, so they update when the data changes. The top of the sheet shows the six key numbers first, followed by the findings and the recommendation, so a manager can read the result without scrolling through tables.

7. What the data shows

Finding	Result
Sales kept	₹32,990 of ₹46,500 total sales (71% of orders completed; 20 cancelled, 27 refunded)
Best city	Pune keeps the most (₹8,450) and loses the least (15%)
Highest on paper	Mumbai (₹10,980) but it ranks #2 once cancelled and refunded orders are removed
Weak spot	Chennai loses 58% of its sales, against 29% for the business overall
Payment	UPI has the highest sales (₹16,210), but Card, Cash and UPI are used fairly evenly
Category	Beverages bring in ₹22,020, about 67% of kept sales

Recommendation: look into why Chennai orders fail (payments, delays or stock) and confirm orders at checkout, and check what Pune does differently.

8. Excel functions used

TRIM, PROPER, IF, AND, COUNTA, SUM, SUMIFS, COUNTIFS, RANK, INDEX, MATCH, MAX, TEXT, conditional formatting, and bar and pie charts.